

ECON 1550 International Finance

# Output and the Exchange Rate in the Short Run

# Behavioral equations in the goods market

Demand for domestic goods

# Equilibrium in the goods market

# A short-run model of the goods market

## Exogenous variables

| Variable | Description           |
|----------|-----------------------|
| $E$      | Nominal exchange rate |
| $I$      | Investment            |
| $G$      | Government spending   |
| $T$      | Taxes                 |
| $P$      | Price level           |
| $P^*$    | Foreign price level   |
| $Y^*$    | Foreign income        |

### Endogenous variables

| Variable | Description               | Equation                                          | Type of equation      |
|----------|---------------------------|---------------------------------------------------|-----------------------|
| $Y$      | Income, production        | $Y = D$                                           | Equilibrium condition |
| $Y_D$    | Disposable income         | $Y_D \equiv Y - T$                                | Identity              |
| EX       | Exports                   | $EX = EX(q, Y^*)_{(+)(+)}$                        | Behavioral            |
| IM       | Imports                   | $IM = IM(q, Y_D)_{(-)(+)}$                        | Behavioral            |
| CA       | Current account           | $CA \equiv EX - IM = CA(q, Y_D, Y^*)_{(+)(-)(+)}$ | Identity              |
| $D$      | Demand for domestic goods | $D \equiv C + I + G + CA$                         | Identity              |
| $A$      | Domestic demand           | $A \equiv C + I + G$                              | Identity              |
| $C$      | Consumption               | $C = C(Y_D)_{(+)}$                                | Behavioral            |
| $q$      | Real exchange rate        | $q \equiv \frac{EP^*}{P}$                         | Identity              |

# DD Curve

$$Y = D$$

$$Y = C + I + G + CA$$

$$Y = C_{(+)}(Y_D) + I + G + CA_{(+)(-)(+)}(q, Y_D, Y^*)$$

$$Y = C_{(+)}(Y_D) + I + G + CA_{(+)(-)(+)}(EP^*/P, Y_D, Y^*)$$

# Short-run FX and money market model

## Exogenous variables

| Variable | Description            |
|----------|------------------------|
| $R^*$    | Foreign interest rate  |
| $E^e$    | Expected exchange rate |
| $Y$      | Real income            |
| $M^s$    | Money supply           |
| $P$      | Price level            |

## Endogenous variables

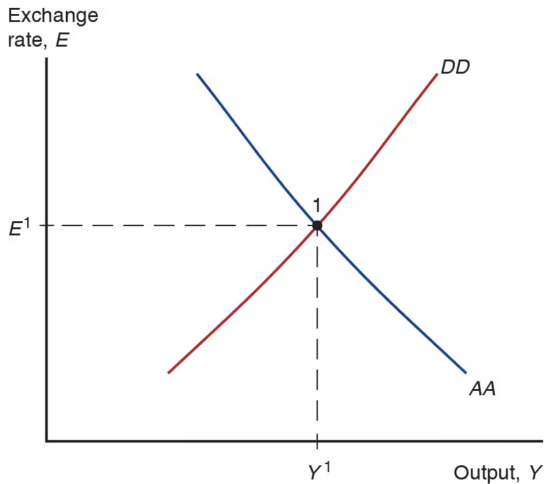
| Variable | Description            | Equation                      | Type of equation      |
|----------|------------------------|-------------------------------|-----------------------|
| $E$      | Exchange rate          | $R = R^* + \frac{E^e - E}{E}$ | Equilibrium condition |
| $R$      | Domestic interest rate | $M^d/P = L(R, Y)$             | Behavioral equation   |
| $M^d$    | Money demand           | $M^d = M^s$                   | Equilibrium condition |

# AA Curve

$$\text{(UIP):} \quad R = R^* + \frac{E^e}{E} - 1$$

$$\text{(MS) = (MD):} \quad \frac{M^s}{P} = L(R, Y)$$

# Short-run Equilibrium in $AA-DD$ model



# Example: DD Schedule

$$C_{(+)}(Y_D) = 1 + 0.75Y_D$$

How to find the DD schedule

$$EX_{(+)}(q, Y_{(+)}^*) = 0.15 + 0.5q + 0.1Y^*$$

$$IM_{(-)}(q, Y_{(+)}^D) = 0.1 - 0.2q + 0.12Y_D$$

$$Y_D = Y - T$$

$$q = \frac{EP^*}{P}$$

# Example: AA Schedule

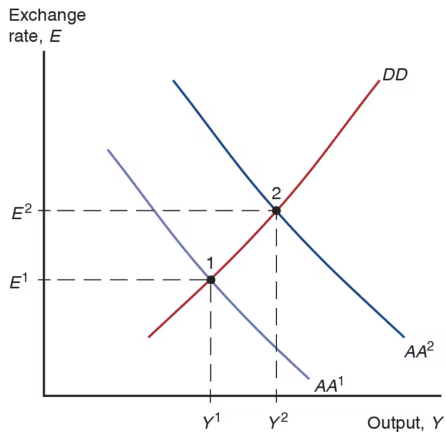
$$R = R^* + \frac{E^e - E}{E}$$

How to find the AA schedule

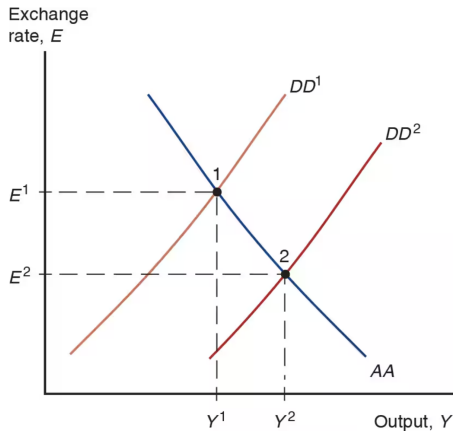
$$\frac{M^d}{P} = \frac{M^s}{P}$$

$$\begin{aligned}\frac{M^d}{P} &= L_{(-)}(R, Y_{(+)}) \\ &= 1.1 - R + 0.05Y\end{aligned}$$

# Temporary Change in Monetary Policy



# Temporary Change in Fiscal Policy



# Full AA-DD Model

DD Schedule:  $Y = C(Y_D) + I + G + CA(EP^*/P, Y_D, Y^*)$

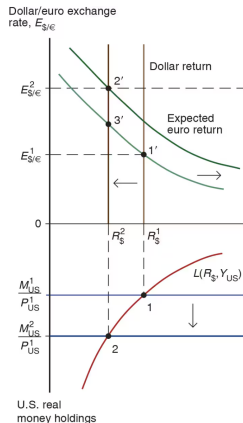
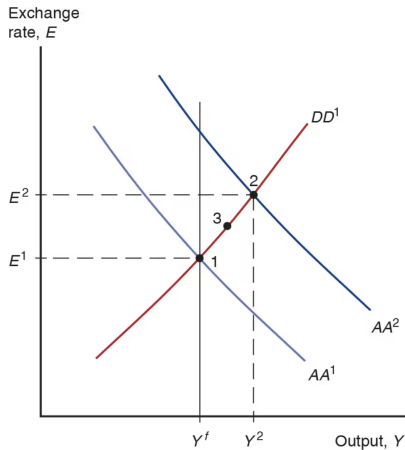
AA Schedule:  $\frac{M^s}{P} = L\left(R^* + \frac{E^e}{E} - 1, Y\right)$

Phillips Curve:  $\pi = \pi^e + \alpha(Y - Y^f)$

Definition of inflation:  $\pi_t = \frac{P_t}{P_{t-1}} - 1$

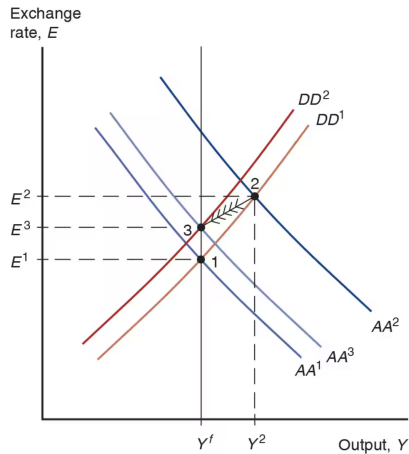
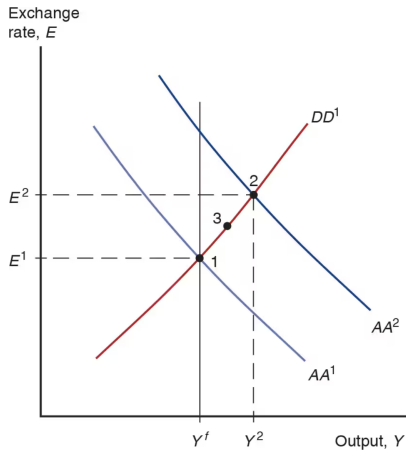
Definition of expected inflation:  $\pi_t^e = \frac{P_{t+1}^e}{P_t} - 1$

# Permanent Shifts in Monetary Policy

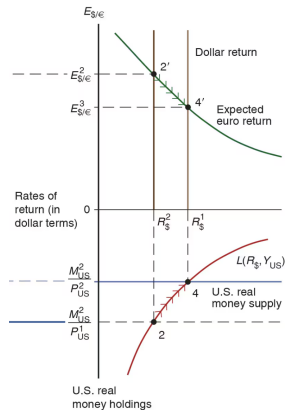
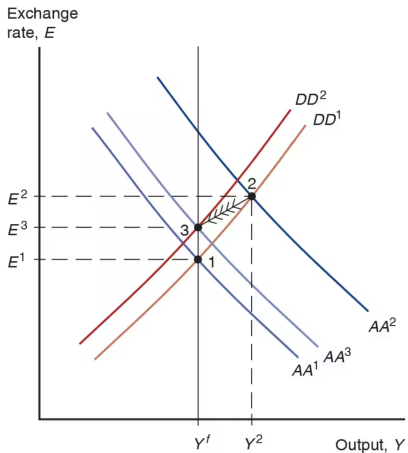


(a) Short-run effects

# Permanent Shifts in Monetary Policy

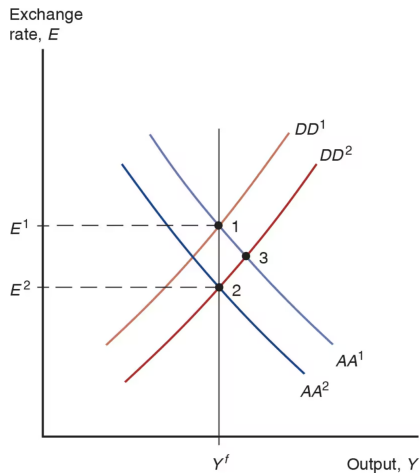


# Permanent Shifts in Monetary Policy



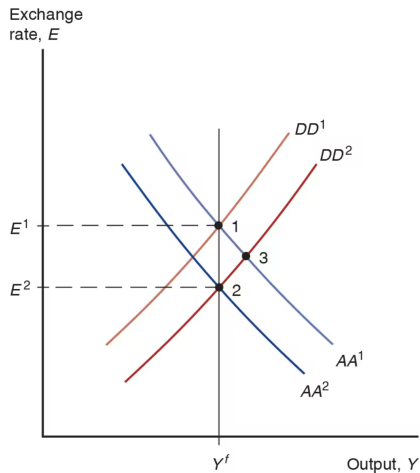
(b) Adjustment to long-run equilibrium

# Permanent Shifts in Fiscal Policy



- Start in a medium run equilibrium at point 1 with  $R_0 = R^*$  and  $E_0^e = E^1$
- Increase in  $G$  shifts  $DD$  to the right
- Since the shift in  $DD$  is permanent,
- At point 3, exchange rate is lower than  $E_1$
- appreciation at point 3 is also expected to be permanent at point 3
- $E^e$  must also go down
- Lower  $E^e$  shifts  $AA$  down
- But how much?

# Permanent Shifts in Fiscal Policy



- There are no more shifts expected to occur,  $E^e$  remains constant
- Therefore  $E = E^e$
- Real money supply is constant
- Real money demand must stay unchanged
- UIP implies  $R$  does not change
- $Y$  must be unchanged as well to keep real money demand unchanged
- If  $Y$  must be equal to its original level  $Y^f$ ,  $AA^1$  must shift to  $AA^2$

# DD Schedule with Tariffs

# Slope of DD Curve with Tariffs